

Project Atlas - Usage-based billing for Northwind. | 8-week Milestone Plan

Today = start of Week 1. Northwind keynote = end of Week 8. Critical-path workstreams are outlined in red.

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Thick red outline = critical path. Critical path: API contract (W1) → Metering + Wallet build (W1-4) → Invoicing (W4-6, at reduced Billing capacity) → Payments/MoR integration (W5-7, gated by partner flag + Tax sign-off) → E2E testing (W6-7) → Go-live (W8). Dev experience runs in parallel and is not critical to Dodo's build, but is critical to Northwind's own integration timeline.

PROJECT ATLAS — Risk & Dependency Register

Ranked by combined likelihood × impact. Reviewed weekly; owner updates status before each leadership check-in.

#	Risk / Dependency	Likelihood & Impact	Early Signal You'd Watch	Mitigation / Plan B	Owner
1	Payment partner pricing flag slips past Week 5	Med / High	No written ETA or status reply from the partner by end of Week 2	Escalate to the partner relationship owner at Week 2, not Week 4. Pre-build a fallback charge path (e.g. a manual/flat residual charge for launch day) that doesn't depend on the new flag, even if less elegant, so keynote day isn't hostage to one vendor.	TPM + Payments lead
2	Billing's 1-engineer leave (W5–6) collides with the hardest workstream (invoicing build)	High / High	Invoicing design/spec isn't fully locked by end of Week 4	Front-load invoicing design and spec review into W3–4, before the engineer leaves, so W5–6 is pure execution against an agreed spec. Scope invoicing to MVP reconciliation only; defer edge cases (credit notes, manual corrections) to a fast-follow.	TPM + Billing lead
3	DB migration overruns past Week 3, further delaying metering ingestion	Med / High	Migration status check-in at end of Week 2 shows <80% complete	Get a hard cutover commitment from the Platform lead in Week 1. If migration slips, pull a Platform engineer off migration support (not off metering) or pull in a second team temporarily to protect the metering deadline.	TPM + Platform lead
4	Tax sign-off incomplete or delayed across 190+ countries	Med / High	Tax hasn't produced a scoped country list / risk-tiering by end of Week 2	Narrow initial launch to countries already covered by Dodo's existing MoR tax rules (likely Northwind's top-revenue markets); sequence remaining countries as a fast-follow after keynote, communicated to Northwind up front as part of scope.	TPM + Compliance/Tax lead
5	Metering/wallet integration surfaces correctness bugs late (double-counting, late-arriving events)	Med / Med-High	Reconciliation discrepancies appear during Week 4 integration testing	Build a reconciliation/audit report into metering from day one (not bolted on later), so mismatches are visible at integration in W4, not discovered inside invoicing in W6 when there's no slack left.	Platform lead

6	Northwind's own integration lags because docs/SDK arrive too late for their engineers	Med / Med	Northwind hasn't kicked off their integration sprint by Week 5	Ship a docs/SDK 'preview' by end of Week 2, based on the frozen API contract, even before the backend is fully built — lets Northwind start in parallel instead of waiting on Dodo's full build.	DevRel lead
7	Unclear ownership at the metering↔wallet handoff between Platform and Billing	Med / Med	Conflicting assumptions surface in standups about who owns deduplication / depletion logic	Write a one-page interface contract (event schema, dedupe ownership, depletion trigger) signed off by both team leads in Week 1, before either team starts building against assumptions.	TPM

PART 1.3 — The Hard Call

Situation at end of Week 3: the payment partner has gone quiet on the pricing flag, and the DB migration slip has left Platform behind on metering. Full scope will not land in 8 weeks.

Section	My call
What I cut	• Day-one global coverage — launch in the markets where Northwind already has the highest revenue concentration and where Dodo's existing rails/tax rules work today, not all 190+ countries. • Strict hourly metering precision at launch — accept near-real-time aggregation for the first invoicing cycle rather than have Platform chase a stricter SLA it's already behind on.
What I defer	• Remaining countries gated by the new partner flag — fast-follow within 2–3 weeks of keynote, with a named owner tracking the flag daily. • Edge-case invoice handling (credit notes, manual corrections, proration) — manual ops review for cycle one; automation follows post-launch.
What I de-risk	• Escalate the partner relationship directly this week — silence past the quoted 2–4 weeks is itself the signal. • Protect the metering deadline above all else, even if it means temporarily borrowing an engineer from DevRel/Platform support. • Get Tax to confirm sign-off is achievable for the narrowed launch-day list specifically, not the full list.
Minimum launchable scope	Customers in the launch-day country list get real, working usage-based billing on keynote day — metering, credit-wallet depletion, low-balance webhooks, and invoicing all live and accurate for that scope. Customers outside that list stay on their current plan until the partner flag lands, then move over within weeks.
The trade-off, stated plainly	Northwind's CEO can truthfully say usage-based billing has launched, but not that it has launched everywhere. We trade a clean, simultaneous global cutover for a credible, working launch on the date that matters, with full coverage close behind. The risk we accept is reputational/narrative, and I'd state that trade-off in exactly those terms to both leadership and Northwind — not soften it.

PART 2 — Technical Fluency

#	Question	Answer
2.1	Why a client-generated event_id, and what breaks without it?	It lets the system recognize “this arrived twice” vs. “this is genuinely new usage.” Without a unique, client-generated ID, a retry or duplicate send gets aggregated as additional usage — the customer’s credit balance depletes faster than reality and their invoice overstates consumption. In billing, that’s an overcharge, not a minor bug.
2.1	Can Northwind safely retry a usage event after a timeout?	Yes — but only if the endpoint is idempotent on event_id, i.e. it checks whether that event_id was already recorded before aggregating, and discards the duplicate if so. The retry is safe because of that property, not because retries are inherently harmless; I’d confirm the dedupe check exists before telling Northwind it’s safe.
2.1	One consequence of metering hourly but invoicing monthly that should be handled deliberately?	Late-arriving or corrected events — usage that shows up after its hour has already been “closed,” or needs to be voided. I’d want an explicit cutoff window per hour (e.g. accept corrections up to 48h later) and a clear rule for what happens after: applied as a next-invoice adjustment, not silently dropped.
2.2	Why does an at-least-once webhook system produce duplicates?	The sender can’t always tell “the receiver never got this” from “the receiver got it but our ack was lost,” and under-delivering a low-balance alert is worse than over-delivering one — so it retries whenever delivery is uncertain. I’d ask Northwind to make their handler idempotent: dedupe on the webhook’s event ID, and design handling to be safe run twice (e.g. “set status to low”, not “decrement a counter”).
2.2	For “sometimes not delivered at all,” what are the first three things to check?	1) Did the event fire on our side — does our delivery log show an attempt when the threshold was crossed? 2) What did their endpoint return on attempts we made — timeout, non-2xx, TLS error — and did we exhaust retries while it was down? 3) Is something between us dropping the request before their app sees it — wrong URL on file, firewall rejecting our IPs, signature verification failing silently?
2.3	What does “metering” actually do, in plain language?	It watches how much of the product each customer uses — here, tokens — and adds it up correctly so we know what they owe.
2.3	Why do credit balances and invoicing depend on metering being correct first?	Both are built directly on the usage number metering produces — balances deplete as usage comes in, invoices charge usage minus credits already applied. If metering miscounts, both are wrong regardless of how well they’re built.
2.3	Why does being Merchant of Record make tax Dodo’s problem, not Northwind’s?	Dodo, not Northwind, is the entity legally selling to each end customer in each country — so charging, collecting, and remitting correct tax in all 190+ countries is Dodo’s legal obligation. A tax mistake is Dodo’s compliance problem.

PART 3 — Stakeholder Communication

Both messages address the situation in Part 1.3 — scope at risk at end of Week 3 — for two different audiences.

Section	Content
3.1 Leadership status update	STATUS: AT RISK Northwind's keynote date itself is not in danger — full scope is. Two things converged this week: the payment partner has gone quiet on the pricing-flag request (originally quoted 2–4 weeks, now Week 3 with no ETA), and the database migration ran long, leaving Platform behind on metering. I'm not waiting for either to resolve before acting. I'm proposing we narrow launch scope: ship metering, credit wallet, and invoicing for the markets/payment methods we can already support without the new partner flag, and treat full 190-country coverage as a fast-follow. This keeps the keynote commitment intact for the substantial majority of Northwind's base. What I need from you: a decision by Friday on whether a phased launch is an acceptable trade, and air cover to raise this with Northwind directly this week.
3.2 Customer message	To: Northwind Technical Lead Quick, transparent update heading into Week 4. Two things have hit our timeline: our payment partner is taking longer than expected on the new pricing flag, and an internal migration pushed our metering build back about a week. Rather than risk the keynote date, we'd like to propose launching live, on day one, for your core markets — full global coverage following within the following weeks. Metering, credit balances, and invoicing will all be live and accurate for that scope; only the rollout sequencing changes. We'll have a finalized country list to you by end of week, and we're happy to walk your team through exactly what's in scope. Let's grab 15 minutes this week to align.
3.3 Reflection	To leadership, I led with the risk itself and asked explicitly for a decision, because their job is to weigh the trade-off and they need the unfiltered picture. To the customer, I led with the plan rather than the cause, because Northwind needs confidence we have a credible path more than our internal account of why it slipped. The leadership update asks for a decision; the customer message offers a meeting — internally I need accountability, externally I need trust.

PART 4 — Process & Ways of Working

4.1 Diagnose & improve: status invisible until something slips

The problem

By Week 3, two independent issues — the partner's silence and the migration overrun — had each been quietly developing for one to two weeks before surfacing together as a combined crisis. Named as "amber" the week each started, escalation and the scope conversation could have begun days earlier, with more options on the table.

The concrete cost

The team loses the days between when a risk actually starts and when it gets named out loud — exactly the days a problem is still cheap to fix.

The change I'd propose

A five-line, async, written weekly status per workstream — status (green/amber/red), what changed, what's blocking, the ask — posted by each owner every Friday, no meeting required. I aggregate it into the leadership update. Deliberately not a status meeting: a 3-engineer team keeps doing 5 minutes of writing; it drops a 30-minute sync.

How I'd know in 4 weeks

Track the callout lag — for any issue that escalated, how many days passed between when it first should have been amber (in hindsight) and when it actually was. Target: that lag trending to zero. Secondary signal: fewer issues jumping straight from invisible to blocking, with no amber step between.

4.2 My operating cadence

Rituals I'd hold

Async daily check-in (2–3 lines/engineer, no call); a 30-min live sync only between the specific team pair with a dependency due that week; the written weekly stoplight update; a standing weekly external-dependency check (partner flag, Tax) logged even when the answer is "no update" — two consecutive "no updates" is itself the escalation trigger.

Rituals I'd skip

A daily standup with the full cross-functional group; a recurring meeting that just reads out what's already written; any retro/process review before the project actually warrants one.

Tracking & surfacing status

One shared, lightweight artifact — the same milestone plan and risk register from Part 1 — updated weekly rather than rebuilt as a separate deck. Leadership and engineers look at the same source of truth.

Protecting velocity

Status is written, decisions are talked — I only call a live meeting for an actual decision or cross-team blocker, never to deliver status already on the page. During crunch weeks, engineers' mornings stay meeting-free by default.